

Goal

Python program that flattens an arbitrarily nested list, classifies elements based on their data type, removes duplicates, sorts each type group, and returns the result as a dictionary

Approach

Step 1:

Created a function called `flatten_typed()` which takes a nested list as input

Step 2:

Created separate sets for:

- integers
- floats
- strings
- booleans

Step 3:

Created a recursive helper function called `flatten()` that traverses the nested list and continues processing until all levels are flattened

Step 4:

Checked whether the current item is a list. If it is a list, each element is processed recursively using the `flatten()` function.

Step 5:

Used `isinstance()` to identify the data type of each element.

Order used:

- bool
- int
- float

- Str

Step 6:

Added each value to its corresponding set removing duplicate values within each data type category

Step 7:

Added exception handling for unsupported data types.

If an element is not one of the supported types, the program raises `TypeError("Unsupported type: <datatype>")`

Step 8:

Created a result dictionary. Each set was converted into a sorted list before being added to the dictionary

Step 9:

Added only non-empty type categories to the final output dictionary

Step 10:

Accepted user input using `input()` and converted it into a Python list using `eval()`

Step 11:

Called the `flatten_typed()` function and displayed the final output.